For general laboratory use.
FOR IN VITRO USE ONLY.

Collagenase A

From Clostridium histolyticum
Clostridiopeptidase A, EC 3.4.24.3 Lyophilizate

Cat. No. 10 103 578 001	100 mg
Cat. No. 10 103 586 001	500 mg
Cat. No. 11 088 793 001	2.5 g

Collagenase B

From Clostridium histolyticum
Clostridiopeptidase A, EC 3.4.24.3 Lyophilizate

Cat. No. 11 088 807 001	100 mg
Cat. No. 11 088 815 001	500 mg
Cat. No. 11 088 831 001	2.5 g

Collagenase D

From Clostridium histolyticum
Clostridiopeptidase A, EC 3.4.24.3 Lyophilizate

Cat. No. 11 088 858 001	100 mg
Cat. No. 11 088 866 001	500 mg
Cat. No. 11 088 882 001	2.5 g

Version August 2004

Store at 2–8°C
strore dry and protected from light!

1. Product overview

Formulation
lyophilized; non-sterile

Preparation
Collagenase A, B and D are prepared from *C. histolyticum* cultures by filtration, ammonium sulfate precipitation, dialysis and lyophilization.

Specific activity
> 0.15 U/mg lyophilizate (collagenase activity):
1 U is the activity which liberates in 1 min at 25°C 1 µmol 4-phenyl-azobenzyl-oxycarbonyl-L-prolyl-L-leucine from 4-phenyl-azobenzyl-oxycarbonyl-L-prolyl-L-leucyl-glycyl-L-prolyl-D-arginine (substrate according to Wünsch) under assay conditions (4).

Additional enzyme activities
The preparations contain other enzyme activities, from which the following are routinely measured for each lot:

Enzyme	Activity
Clostripain	1 U catalyzes the hydrolysis of 1 µmol N-Q-benzoyl-L-arginine ethylester (BAEE) per min at 25°C and pH 7.6 after activation with 1 mM calcium acetate and 2.5 mM dithiothreitol.
Tryptic activity	With BAEE as substrate: 1 U is that enzyme activity which hydrolyzes 1 µmol BAEE in 1 min at 25°C and pH 7.6.
Protease activity	1 U is that protease activity which is causing an absorption increase of 0.001 in 1 min at 25°C in the standard azocoll test.

Collagenase A, B and D contain different ratios of the various proteolytic activities:

Collagenase A	Normal balanced ratio of enzyme activities.
Collagenase B	Normal to high collagenase activity and higher than normal clostripain activity (usually >10 U/mg).
Collagenase D	Normal to high collagenase activity and very low tryptic activity (usually <0.1 U/mg).

Inhibitors
EDTA, EGTA, Cys, His, DTT, 2-mercaptoethanol.
Note: Collagenase is not inhibited by serum

Activators
Ca^{2+}

pH Optimum
6.0–8.0.

Application
Collagenase from C. histolyticum is now widely used for the disaggregation of all kind of tissues (e.g. lung, heart, muscle, bone, adipose tissue, liver, kidney, cartilage, mammary gland, placentae, blood vessels, brain, all kind of tumors) and for the preparation of single cell suspensions for the establishment of primary cell culture systems.
Clostridium collagenase from Roche has been used to prepare cells from many types of tissue, e.g. hepatocytes, adipocytes, pancreatic islets, epithelial cells, muscle cells, endothelial cells etc. (5-19). However, suitability of each lot of the enzyme for disruption of a particular tissue should be determined empirically.

Reconstitution
In any balanced salt solution (e.g. HBSS) (see table 1).
Tab. 1: Composition of selected balanced salt solutions[1].

	Ringer[2]	Tyrode[3, 4]	Gey[5]	Earle[6]	Puck[7]	Hanks[8]	Dulbecco (PBS)[9, 10]
NaCl	9.00	8.00	7.00	6.80	8.00	8.00	8.00
KCl	0.42	0.20	0.37	0.40	0.40	0.40	0.20
$CaCl_2$	0.25	0.20	0.17	0.20	0.012	0.14	0.10
$MgC_2 \times 6\,H_2O$		0.10	0.21			0.10	0.10
$MgSO_4 \times 7\,H_2O$			0.07	0.10	0.154	0.10	
$Na_2HPO_4 \times 12\,H_2O$			0.30		0.39	0.12	2.31
$NaH_2PO_4 \times H_2O$		0.05		0.125			
KH_2PO_4			0.03		0.15	0.06	0.20
$NaHCO_3$		1.00	2.27	2.20		0.35	
Glucose		1.00	1.00	1.00	1.10	1.00	
Phenol ret				0.05	0.005	0.02	
Atmosphere	air	air	95% air/ 5% CO_2	95% air/ 5% CO_2	air	air	air

1 Amounts are given as grams per liter of solution.
2 Ringer, S. (1985) J. Physiol. (London) **18**, 425.
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5 Gey, G. O. & Gey, M. K. (1936) Am. *J. Cancer* **27**, 55.
6 Earle, W. R. (1943) *J. Natl. Cancer Inst.* **4**, 165.
7 Puck, T. T., Cieciura, S. J. & Robinson, A. (1958) *J. Exp. Med.* **108**, 945.
8 Hanks, J. H. & Wallace, R. E. (1949) *Proc. Soc. Exp. Biol. Med.* **71**, 196.
9 PBS, phosphate-buffered saline
10 Dulbecco, R. & Vogt, M. (1954) *J. Exp. Med.* **99**, 167.

Roche

Working concentration

approx. 1 mg/ml (0.1%, w/v).

Storage/Stability

The lyophilizate is stable at 2–8°C, when stored dry and protected from light, until the expiration date printed on the label.
The reconstituted solution is stable at −15 to −25°C.

Background information

Bacterial collagenase, or more accurately clostridiopeptidase A, is a protease with a specificity for the X-Gly bond in the sequence Pro-X-Gly-Pro, where X is most frequently a neutral amino acid. Such sequences are found in high frequency in collagen, but only rarely in other proteins. While many proteases can hydrolyze single-stranded, denatured collagen polypeptides, clostridiopeptidase A is unique among proteases in its ability to attack and degrade the triple-helical native collagen fibrils commonly found in connective tissue (1–3).
Purified clostridiopeptidase A alone is usually inefficient in dissociating tissues due to incomplete hydrolysis of all collagenous polypeptides and its limited activity against the high concentrations of non-collagen proteins and other macromolecules found in the extracellular matrix.
The collagenase most commonly used for tissue dissociation is a crude preparation from C. histolyticum containing clostridiopeptidase A in addition to a number of other proteases, polysaccharidases and lipases.
Crude collagenase is apparently ideally suited for tissue dissociation since it contains the enzyme required to attack native collagen, in addition to the enzymes which hydrolyze the other proteins, polysaccharides and lipids in the extracellular matrix of tissues.
Collagenase A, B and D are prepared from the extracellular culture filtrate of Clostridium histolyticum. These crude preparations contain collagenase and other proteases, including clostripain, a trypsin-like activity and a neutral protease. This mixture of enzyme activities makes crude collagenases ideally suited for gentle dissociation of tissue to generate single cells. Collagenase A, B and D contain different ratios of the various proteolytic activities. This allows for selection of the preparation best suited for disaggregation of a particular tissue.

Unit definition

Collagenase from Roche is assayed in Wünsch units (1µmol of product formed per minute at 25°C with Wünsch substrate [4]).
Frequently, collagenase activities are given in Mandl units (1 µmol leucine liberated from collagen in 5 h at 37°C).
Unfortunately, there is no consistent conversion factor between the two units of activity, since the Mandl unit depends, in part, on the concentration of contaminating proteases in the collagenase preparation, an indefinable variable. A purer collagenase preparation would actually give a lower specific activity in Mandl units than a crude preparation. Clostridium preparations typically give conversion factors of approx. 1:1800 (e.g. a particular lot of Clostridium collagenase contained approx. 0.15 Wünsch U/mg and 250 Mandl U/mg).

2. Procedures and required materials

General

Two types of procedures are commonly used. The first involves mincing tissue and incubating the pieces in a collagenase solution with mild agitation.
Cells are gradually released from the tissue during the collagenase treatment. The second involves perfusing an organ with the collagenase solution. Cells are gradually released into the perfusate or the tissue is then dissociated by mild mechanical treatment.

Additional reagents and material required

- PBS, sterile
  or another balanced salt solution
- Filter membrane, 0.22 µm
- Nylon mesh or gaze

Protocol

The following working instruction describes, as an example, a procedure for minced tissue:

Step	Action
1	Dissolve the non-sterile, lyophilized enzyme in a balanced salt solution and filter sterilize through a 0.22 µm filter membrane.
2	Wash the tissue in sterile PBS or another balanced salt solution.
3	Remove undesirable tissue like fat or necrotic material and cut the remaining tissue with a scalpel into 1-3 mm cubes.
4	Add collagenase solution [usually 0.1% to 0.25% (w/v)]. ***Note***: It is possible, but in most cases not necessary, to add other enzymes such as pronase*, hyaluronidase*, elastase* or trypsin* to the collagenase solution.
5	Incubate at 37°C until disaggregation is complete.
6	Check for effective disaggregation. If the cell suspension becomes viscous due to DNA release from digested cells, add DNase I* to alleviate this problem. If necessary separate undissociated fragments from single cells by collecting the supernatant after allowing the fragments to settle and add fresh enzyme solution to the tissue fragments. The cell suspension can be passed through a nylon mesh or gaze to remove any undigested fragments.
7	Centrifuge the supernatant(s) at 50–100 × g for about 3 min.
8	Resuspend the pellet in medium and seed as usual. ***Note***: For organ perfusion procedures special products (collagenase H* for hepatocyte isolation, collagenase P* for pancreatic islet isolation) are available.

3. References

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4. Ordering Guide

For further information please access our web-site address at: http://www.roche-applied-science.com

Product	Pack Size	Cat. No.
Collagenase H	100 mg 500 mg 2.5 g	11 074 032 001 11 074 059 001 11 087 789 001
Collagenase P	100 mg 500 mg 2.5 g	11 213 857 001 11 213 865 001 11 213 873 001
Collagenase/Dispase	100 mg 500 mg	10 269 638 001 11 097 113 001
Pronase	1 g 5 g	10 165 921 001 11 459 643 001
DNase I	100 mg sterile	11 284 932 001
Hyaluronidase	100 mg	10 106 500 001
Elastase	10 mg 50 mg	11 027 891 001 11 027 905 001
Trypsin	500 mg 2 g	10 109 819 001 10 109 827 001
Trypsin Solution	100 ml sterile	10 210 234 001
Buffers in a Box, pre-mixed PBS Buffer, 10×	4 l	11 666 789 001

* available from Roche Applied Science

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